Do Better Generators Make Worse Teachers? Disentangling Diversity, Label Alignment and Fidelity in Synthetic Training Data

Anonymous CVPR submission

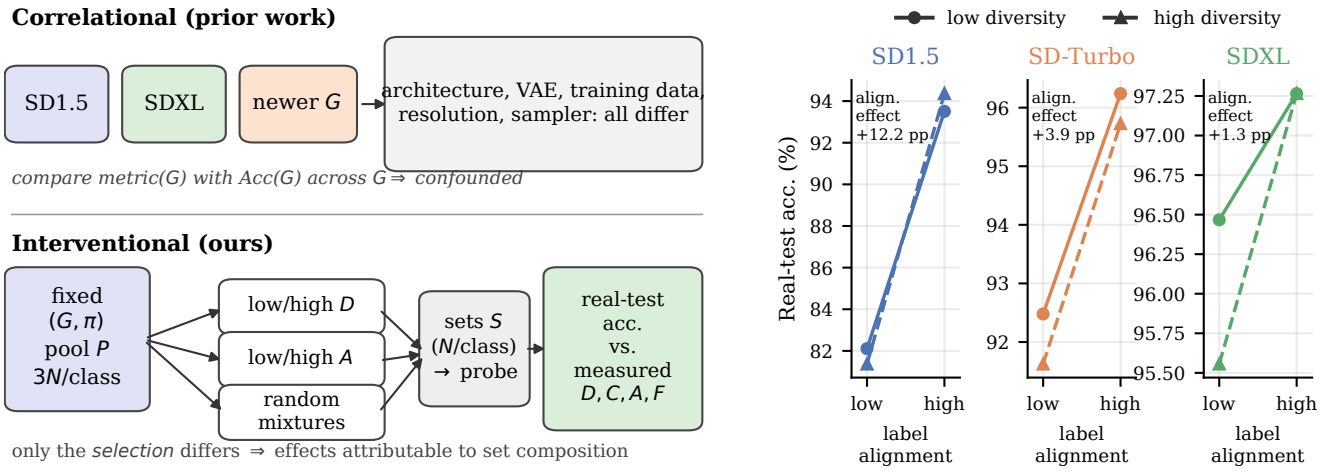


Figure 1. Left: cross-generator comparisons change everything at once; we instead fix the generator G and prompt regime π and vary only which images form the training set. Right: a 2×2 intervention on 20 CUB-200 classes. Raising label alignment lifts real-test accuracy by +12.2, +3.9 and +1.3 points for SD1.5, SD-Turbo and SDXL; raising diversity does not help (−0.7 to +0.1). The fixable deficit of synthetic data is correctness, and it shrinks as generators improve. Note the per-panel y-scales.

Abstract

Newer text-to-image generators produce images that look better yet train worse classifiers, a regression recently attributed to a collapse toward narrow, aesthetic-centric distributions. That diagnosis is correlational, rests on generic ImageNet classes, and comes without a remedy. We ask which property of a synthetic training set actually drives real-test accuracy, and we answer by intervention rather than correlation. Holding the generator, prompt and pool of images fixed, we construct training sets that differ in only one measured property: intrinsic diversity, coverage of the real distribution, label alignment, or per-sample fidelity. We use a 2×2 factorial and a balanced random-subset design with a Shapley decomposition. Under a hard compute limit, this version is a pilot: three open generators (SD1.5, SD-Turbo, SDXL) on 20 classes of the fine-grained CUB-200-2011 dataset, evaluated with a frozen-backbone linear probe. In this setting the regression does not replicate: SDXL is the best teacher (96.2% vs. 91.5% and 90.9% under class-name prompts; real data 98.6%). Within a fixed pool, label alignment moves accuracy (+12.2, +3.9 and +1.3 points for SD1.5, SD-Turbo and SDXL) while diversity has no measurable effect, but the four properties are too collinear for a reliable Shapley attribution (within-pool R^2=3.6%). DAS, a training-free recipe (oversample, filter by label alignment, then select for coverage), closes 25–44% of the gap between naive synthetic and matched real data, but a plain alignment filter does about as well. We report where it does not help.

1. Introduction

Text-to-image (T2I) models are an appealing source of training data. They are cheap to query, free of collection cost, and can in principle be asked for any class in

any condition [2, 13, 29, 30]. It is natural to expect that as generators improve, the data they produce should improve with them.

Adamkiewicz et al. [1] showed that the opposite happens. Across thirteen open generators released between 2022 and 2025, classifiers trained only on synthetic ImageNet images and tested on real ones get steadily worse with newer models, even as prompt following and visual quality improve. Their analysis links the regression to degraded texture and high-frequency statistics and to a collapse of the synthetic distribution: high density, low coverage.

This is an important observation, but it leaves the mechanism open in three ways that matter for anyone who wants to use synthetic data. First, the evidence is correlational. Generators differ in architecture, autoencoder, training data, default resolution and guidance behaviour all at once. A cross-generator association between coverage and accuracy cannot separate the effect of coverage from everything else that changed. Second, the evidence concerns 200 generic ImageNet classes that generators have seen countless times. In fine-grained domains such as bird species or aircraft variants, a generator may render a beautiful, diverse image of the wrong species. In that case label alignment, not diversity, could be the binding constraint. Third, no remedy is offered short of obtaining detailed captions of real images, which presupposes the real data that synthetic data was meant to replace.

We address these gaps with a simple change of method: we intervene on the training set instead of comparing generators. For a fixed generator and prompt regime we draw a large pool of images and then construct many training sets of identical size from that pool. These sets differ systematically in one measured property while the others are held fixed or balanced. Because generator, prompt, autoencoder and resolution are now constant, differences in real-test accuracy can be attributed to the manipulated property of the set rather than to the model that produced it. We use two designs. One is a 2×2 factorial over diversity and alignment. The other is a balanced random-subset design whose variation supports a Shapley decomposition of explained variance [11, 21].

Taking this route forces several distinctions that correlational studies tend to blur, and we make them explicit. Intrinsic diversity (how spread out a set is, measured by the Vendi score [8]) is not the same as coverage (how much of the real distribution the set reaches [18, 24]). A generator can be highly diverse along axes that do not matter, such as artistic style. Label alignment (does the image show the prompted class?) is not prompt following in the GenEval sense, and the predecessor found the latter inversely related to utility [1]. Fidelity measured per sample (precision, density) is not FID, which conflates fidelity and diversity in a single distance [14, 19].

The decomposition suggests a remedy that needs no generator training. DAS (Diversity-matched, Alignment-filtered Sampling) oversamples each class, discards images that a few-shot real classifier or a text-only scorer does not recognise as the prompted class, and then selects for coverage with greedy k-center [10, 32]. We compare it with natural alternatives, including simply training on the whole oversampled pool, so that the benefit of selection is not confused with the benefit of more images.

Findings. On 20 CUB-200-2011 classes with SD1.5, SD-Turbo and SDXL:

- No generational regression here. SDXL trains the best classifier, and of the standard generator metrics only $\mathrm{FD}_{\mathrm{DINO}}$ ranks the three generators as their data does; precision ranks the worst teacher first (Sec. 6.1).
- Correct before diverse. Holding the pool fixed, raising label alignment lifts accuracy by up to 12.2 points; raising diversity does not help (Fig. 1).
- The gap lives in a few classes the generator does not know (Fig. 4).
- DAS, a training-free selection rule, beats random selection in every setting and all $3\times$ images in 5/6, but most of its gain comes from the filter (Fig. 5).

We state our hypotheses in advance (Sec. 3.4) and report negative outcomes as findings.

2. Related work

Training on generated images. Early work established that diffusion-generated images can train or pretrain useful classifiers. These include ImageNet clones from Stable Diffusion [30], synthetic data for zero- and few-shot recognition [13], fine-tuned Imagen data that improves ImageNet accuracy [2], and synthetic contrastive pretraining [38, 39]. Fan et al. [7] characterised scaling laws for synthetic images and their limits. Geng et al. [9] showed that retrieving real images from a generator's own training set beats generating from it. Singh et al. [34] benchmarked the robustness of synthetic-trained models. Most of this work uses Stable Diffusion 1.x/2.x [29]. Adamkiewicz et al. [1] extended the comparison to thirteen generators and found a regression with newer models, which is the starting point of this paper. Their analysis is correlational across generators on generic classes; ours intervenes within generators on fine-grained classes and proposes a remedy.

Diversity, prompts and augmentation. Several works observe that synthetic data lacks diversity and attack it at the prompt: by diversifying prompts [33], using captions as prompts [20], or editing real images toward new domains [5, 40]. For fine-grained classes, SaSPA [23] conditions generation on structure to preserve subject identity. DataDream [16] fine-tunes the generator on a few real shots. Real-Fake [43] frames synthesis as distribution matching. These methods change the generator or its conditioning. DAS is complementary: it leaves the generator untouched and changes only which samples are kept, so it can be stacked on top of any of them.

Data selection and label noise. Selecting a coreset by greedy k-center [10, 32] and pruning data by difficulty [35] are established for real data. Filtering by a classifier's agreement with the given label is a standard tool against label noise [25]. In synthetic data a misaligned image is a form of label noise whose rate depends on how well the generator knows the class. We study how filtering and coverage selection interact when the pool itself is collapsed.

Evaluating generative models. FID [14] and KID [3] measure a distance between distributions and therefore mix fidelity with diversity. Inception features are also biased toward ImageNet classes [19]. Precision/recall [18] and density/coverage [24] separate the two. Stein et al. [37] recommend DINOv2 features [26] over Inception. The Vendi score [8] measures diversity without a reference set. We use these metrics as candidate explanatory variables and test which ones predict training utility under intervention.

3. Setup, measurements and hypotheses

Notation. A task has K classes, a real training set D_r, and a real test set D_{te} that is never used for anything except final evaluation. A generator G with prompt regime π produces a pool $P_{G,\pi}$ of mN images per class (m=3). A training set $S \subset P$ holds exactly N images per class. The test accuracy of a classifier trained on S is $\text{Acc}(S)$. We fix N to the per-class size of D_r so that synthetic and real sets have equal size.

We carve a small alignment split $A \subset D_r$ of k=5 images per class. It is used only to score label alignment and as the training set of the matched real 5-shot baseline. Any method that uses A is therefore compared against a classifier that is given the same real images.

3.1. Data, generators and prompts

Datasets. CUB-200-2011 [42] (200 bird species, about 30 training images per class) is fine-grained: classes are distinguished by local cues such as plumage patterns, and they are rarer in web captions than ImageNet categories. In this version we use 20 classes at evenly spaced class indices (no hand-picking), with N=24 real training images per class after carving A and 585 real test images in total (one test image is 0.17 points). FGVC-Aircraft [22] and the exploratory HAM10000 domain [41] belong to the full design and were not run.

Generators. The full design uses four open models spanning three generations: SD1.5 [29], SDXL [27], SD3.5 [6, 36] and FLUX.1-dev [4]. SD3.5 and FLUX.1-dev do not fit on the free 15 GB T4 GPUs we had, so this version runs SD1.5, SDXL, and SD-Turbo, an adversarially distilled model [31] used as a cheap third generator. SD-Turbo is not a "next version" of SD1.5. We use 25 sampling steps and classifier-free guidance [15] of 2.0 for SD1.5 and SDXL, following the low-guidance setting known to favour diversity [1, 30]; SD-Turbo uses 4 steps and no guidance, as it is distilled. SD1.5 and SD-Turbo are generated at 512^2. SDXL is generated at 768^2 rather than its native 1024^2, which was necessary to fit memory and the session time limit. Resolution, steps and guidance are therefore confounded with the generator, and we report that rather than hide it. All images are resized to 224^2 before training.

Prompt regimes. P1 uses the class name with a domain hint ("a photo of a {class}, a type of bird"). P2 uses LLM-diversified templates that vary pose, background, lighting, season and viewpoint, with the class name always present; the templates are written once per domain without access to real images. P3 (detailed captions of real images from a vision-language model) is an oracle regime that leaks information about the real distribution [1]; it was not run in this version. P1 and P2 are run for all three generators. Each pool holds mN=72 images per class (1,440 images), and equal-size training sets use 480 images.

3.2. Measured properties

All metrics are computed per class and averaged, at equal N, because several of them (notably Vendi) depend on sample size. The primary embedding is DINOv2 ViT-B/14 [26], which is also the backbone of the main learner. (Recomputing all metrics in CLIP ViT-L/14 [28] space as a robustness check is part of the design but was not run here.)

Diversity (reference-free). The Vendi score [8]: the exponential of the entropy of the eigenvalues of $\frac{1}{N}K$, with

K the cosine-similarity kernel of the N embeddings. It is the effective number of distinct samples, and it does not ask whether those samples resemble real data.

Coverage (reference-based). The fraction of real training images whose k-NN ball (k=5) contains at least one synthetic image [24], and improved recall [18] (k=3, given only 24 images per class). These ask whether the synthetic set reaches the real distribution. The distinction from Vendi is central to our analysis. A set can raise Vendi by spreading into regions real data never occupies, for example painterly styles, without raising coverage at all.

Label alignment. Two independent scorers, both outputting $a(x, y) \in [0, 1]$ for image x prompted with class y. A_{real} is the softmax probability of y under a nearest-class-mean classifier built from the alignment split A in CLIP image space. A_{text} is CLIP zero-shot probability over the K class names and uses no real images at all. Both are computed in CLIP space, never in the DINOv2 space of the learner, to avoid scoring images with the same features that are later trained on.

Fidelity (per sample). Improved precision [18] and density [24]: how many synthetic samples land on the real manifold. We also report FD in DINOv2 space (FD_{DINO} [37]). Because FD-type metrics mix fidelity and diversity, we treat them as proxies to be tested, not as one of the factors.

3.3. Learners

Main: frozen DINOv2 linear probe. Multinomial logistic regression on frozen, ℓ_2-normalised DINOv2 ViT-B/14 [CLS] features [17]. Weight decay is selected on a held-out synthetic validation split. Selecting on real validation data would leak real information into a nominally synthetic-only pipeline. The probe measures images as supervision for a head, not as pretraining data, and is fast enough to train thousands of times, as the interventional designs require.

Control: from-scratch ResNet. A ResNet-18 [12] trained from random initialisation is part of the design, to check whether it ranks training sets the same way the probe does. It is implemented but was not run in this version, so our claims are confined to the supervision setting.

Baselines and confound controls. Real-only training at $k \in \{5, 10, \text{all}\}$ images per class. Autoencoder ceiling: we pass D_r through a generator's VAE (encode then decode, no diffusion) and train on the reconstructions. This bounds how much of the gap is due to latent-space artifacts alone [1]. We ran it for SD1.5 and SD-Turbo, not for SDXL (full-resolution decoding did not fit in memory). Contamination: generators and DINOv2 may have seen these benchmarks' images; we cannot rule this out, and it would inflate synthetic and probe accuracy alike, biasing toward synthetic data.

3.4. Hypotheses

We fixed these hypotheses before running the interventional experiments.

H1. Within a fixed pool, coverage and label alignment explain more variance in real-test accuracy than per-sample fidelity, and reference-free diversity (Vendi) explains less than coverage.

H2. Across generators, newer models score higher on fidelity and lower on coverage under P1, so FID and precision mis-rank generators as data sources.

H3. The gap is larger for classes the generator knows poorly. We test this per class by regressing the class-level gap on class familiarity, the mean A_{text} of the class under P1.

H4. Alignment filtering followed by coverage selection (DAS) closes a meaningful share of the gap. We commit to reporting the share however small, and to comparing against training on the full oversampled pool.

4. From correlation to intervention

Comparing generators confounds every property of the data with every property of the model. We instead fix (G, π), and with it the autoencoder, resolution, sampler and training corpus, and vary only which N images per class are kept from the pool $P_{G,\pi}$. The manipulated variable is then the composition of the training set, and its effect on Acc is identified up to properties of the selected images that we did not measure (Sec. 7). All selection steps use CLIP embeddings. All diversity, coverage and fidelity measurements use DINOv2 embeddings. Selection therefore cannot trivially maximise the quantity it is scored on.

Diversity arm. Fix an alignment threshold τ (the pool's 40th percentile of A_{real}) and restrict to images above it. From the surviving set, choose N per class in one of three ways. Clustered takes the N nearest neighbours of a random seed image, giving low diversity. Random gives the pool's natural diversity. k-center uses greedy farthest-point selection [10], giving

high diversity. Mean alignment is held fixed by construction, up to within-threshold variation. k-center is known to favour outliers, which may have lower fidelity, so we measure precision in every arm.

Alignment arm. Fix the selection rule to random and vary the alignment quantile from which images are drawn: bottom, middle or top third of A_{real}. High-alignment images tend to be prototypical and therefore less diverse, so this arm alone does not hold diversity fixed. We correct this by matching: within each tercile we draw 200 random candidate subsets and keep the one whose Vendi score is closest to the pool median.

2×2 factorial. Crossing low/high diversity with low/high alignment, constructed as above, gives four training sets per (G, π). With three selection seeds and three training seeds each, the design estimates both main effects and their interaction. An interaction is plausible a priori: diversity may help only when images are correctly labelled, since spreading out misaligned images adds varied label noise.

Balanced random-subset design and attribution. Factorial cells test two factors at two levels. To attribute variance across all four candidate factors, we additionally draw R=150 subsets per (G, π) (the full design uses 500). Each is formed by mixing random, k-center, clustered and quantile-restricted draws in random proportions. This produces broad variation in diversity D, coverage C, alignment A and fidelity F (we report their correlation structure). We fit

$$\text{Acc}(S) = \beta_D \tilde{D} + \beta_C \tilde{C} + \beta_A \tilde{A} + \beta_F \tilde{F} + \gamma_{G,\pi} + \varepsilon, \quad (1)$$

with standardised regressors and a fixed effect γ for each generator–prompt cell. Variance is therefore explained only within pools. We report the Shapley (LMG) share of R^2 for each factor [11, 21]: its average marginal contribution over all orderings of regressors. Unlike standardised coefficients, these shares remain interpretable when regressors are correlated. We also fit the model between cells, without fixed effects, over the generator–prompt configurations; it has very few points and no control of confounds, so it is descriptive.

5. DAS: Diversity-matched, Alignment-filtered Sampling

DAS turns the decomposition into a recipe that needs no gradient step on the generator. For each class y:

1. Oversample. Generate mN images (m=3) under P2.
2. Filter. Discard images with $a(x, y) < \tau$. Here τ is a pool quantile chosen on the synthetic validation split, never on real test data (we use the 40th percentile). If fewer than N images survive, keep the top-N by alignment.
3. Select. From the survivors, choose N images by greedy k-center in CLIP space [10, 32]. The first centre is the survivor with the highest alignment, so that the most reliable image anchors the set.

DAS-real uses A_{real} and therefore the 5-shot alignment split. DAS-text uses A_{text} and no real image at any stage. Selection costs $O(mN^2)$ distance evaluations per class, which is negligible next to generation.

What DAS must beat. Random-N; All-mN, the entire pool, with three times the images at the same generation cost, which separates the value of selecting from the value of having more; Filter-only and k-center-only; Real 5-shot, the same real images DAS-real sees, used directly; and the hybrid Real 5-shot + DAS. We summarise each method by the fraction of gap closed,

$$\Delta = \frac{\text{Acc(method)} - \text{Acc(Random-}N\text{)}}{\text{Acc(Real-}N\text{)} - \text{Acc(Random-}N\text{)}}, \quad (2)$$

undefined when the denominator is near zero, in which case we report raw accuracies only. Δ>1 would mean the method surpasses real data of equal size.

6. Results

All results are for 20 CUB classes, one generation seed per pool, and a 585-image test set; one test image is 0.17 points. Intervals reflect selection and training randomness, not generation randomness.

6.1. Cross-generator picture

Figure 2 and Tab. 1 report the uncontrolled comparison that motivates the paper. It tests, at small scale and on fine-grained data, whether the experiment of [1] replicates.

The regression does not replicate here. Under P1, accuracy is 90.9% (SD1.5), 96.2% (SDXL) and 91.5% (SD-Turbo), ordered by release date; Kendall's τ over release order is +0.33 (P2: +0.33), so there is no monotone decline. The newest large model, SDXL, is the best teacher, 96.2% against 98.6% for real data. SD-Turbo is no better than SD1.5 under P1 (91.5% vs. 90.9%, about three test images) but is better under P2 (93.8% vs. 90.6%). We read this as a finding about domain dependence and scale: with three generators we cannot say the regression is absent in general.

Precision and coverage do not track accuracy (H2). Precision falls from SD1.5 (0.62) to SDXL (0.27) to SD-Turbo (0.03), and coverage falls the same way (0.70, 0.36, 0.03), while SDXL has the highest accuracy. Higher fidelity for newer models, as H2 predicted, is therefore not observed; coverage does fall for newer models, but it does not drive accuracy. The ranking by $\mathrm{FD}_{\mathrm{DINO}}$ (SDXL 0.74, SD-Turbo 0.81, SD1.5 0.93; lower is better) agrees with the accuracy ranking, whereas precision ranks SD1.5, the worst teacher, first. This is the one place the "looks better, teaches worse" pattern appears, and it rests on a measure computed from only 24 images per class (recall is near zero, 0.013–0.040, so precision/recall values are noisy).

Prompts and the autoencoder ceiling. P2 changes accuracy by less than 0.4 points for SD1.5 and SDXL and raises SD-Turbo from 91.5% to 93.8%; P3 was not run. Re-rendering real images through the autoencoder gives 98.3% (SD1.5) and 98.5% (SD-Turbo), against 98.6% for the originals, so autoencoder artifacts explain at most 0.3 points of the gap; the rest is a property of what the generators draw.

6.2. Interventional decomposition

Factorial (Fig. 1, Tab. 2). Raising label alignment helps a lot for the weakest generator (+12.2 points for SD1.5), less for SD-Turbo (+3.9) and little for SDXL (+1.3). Raising diversity does nothing or slightly hurts (+0.1, -0.7, -0.5). The interaction is positive for all three (+1.6, +0.3, +0.9), which is the direction the label-noise argument of Sec. 4 predicts (diversity helps, or hurts less, when images are correctly labelled), but its size is below one point for two of the three generators, so we do not claim more. Plausibly, weaker generators more often draw the wrong or an unrecognisable bird, so filtering helps most there (untested).

Within-pool decomposition (H1). Over 150 random subsets per generator, the regression explains only 3.6% of the variance in accuracy. The Shapley shares are D 0.5%, C 1.6%, A 1.0% and F 0.5%. Numerically $C{+}A > F$ and $C > D$, as H1 predicts, but the totals are tiny and we computed no confidence intervals, so we neither confirm nor refute H1. The factors are also strongly collinear: Vendi correlates 0.91 with coverage and 0.92 with precision, and variance inflation factors range from 36 to 147 (values above 10 are usually taken as serious), which limits what any decomposition can claim. The between-generator version has only three points for four predictors and is not identified, so the paper's central contrast, cross-generator correlation versus within-pool attribution, cannot be drawn at this scale. Recomputation in CLIP space and with A_{text} in place of A_{real} was not run.

Class familiarity (H3). Regressing the per-class gap on familiarity gives negative slopes for all three generators (SD1.5 -0.29 [-0.52, -0.05], SD-Turbo -0.95 [-1.32, 0.23], SDXL -0.18 [-0.24, 0.25]; 95% bootstrap intervals), i.e. a larger gap for classes the generator knows poorly. The OLS fits are strong for SD-Turbo (R^2=0.58, p=0.0001) and SDXL (R^2=0.49, p=0.0006) and weak for SD1.5 (R^2=0.16, p=0.0854), but the bootstrap intervals for SD-Turbo and SDXL include zero while SD1.5's does not, because the per-class gaps are zero-inflated. The two tests disagree, so H3 is only weakly supported. The least familiar class is the same for all three generators, Chuck-will's-widow (familiarity 0.11, 0.25, 0.20; gaps of 23.1%, 100.0% and 19.2%); for SD-Turbo the probe classifies none of its test images correctly. We did not investigate why.

6.3. From-scratch control and DAS

From-scratch control. Not run in this version, so conclusions are restricted to the supervision setting (a head on a real-pretrained backbone); we make no claim that the ranking carries over to pretraining.

DAS against the baselines (Tab. 3). DAS-real improves over random selection by +2.4 (SD1.5), +2.1 (SD-Turbo) and +1.2 points (SDXL), and DAS-text by +3.4, +1.2 and +1.2. In terms of Δ, DAS-real closes 0.30, 0.43 and 0.44 of the gap and DAS-text 0.43, 0.25 and 0.44. DAS beats training on all $3N$ images in five of six comparisons, so the gain is not just from having more images (SD-Turbo with DAS-text, 95.0%, is below All-mN, 95.4%). But filtering alone is about as good: it equals or beats DAS in three of six cases (SD-Turbo with A_{text}, SDXL with both scorers), so we cannot credit DAS's diversity step with an improvement. k-center alone is below random for SD1.5 and SD-Turbo. Real 5-shot (97.8%) still beats every synthetic-only set, so five real images are worth more than a curated synthetic set. Adding DAS-selected images to them gives 97.9%, 98.5% and 97.6%, within about 0.7 points of 5-shot alone, so the hybrid is not clearly useful at this scale.

Ablations. Moving the alignment threshold over the 20th, 40th and 60th percentile changes DAS accuracy by about one point at most (SD1.5 93.0–93.3%, SD-Turbo 95.0–96.1%, SDXL 96.4–97.4%).

Mixing with real data. With 5 or 10 real images per class, adding synthetic images changes accuracy by be-

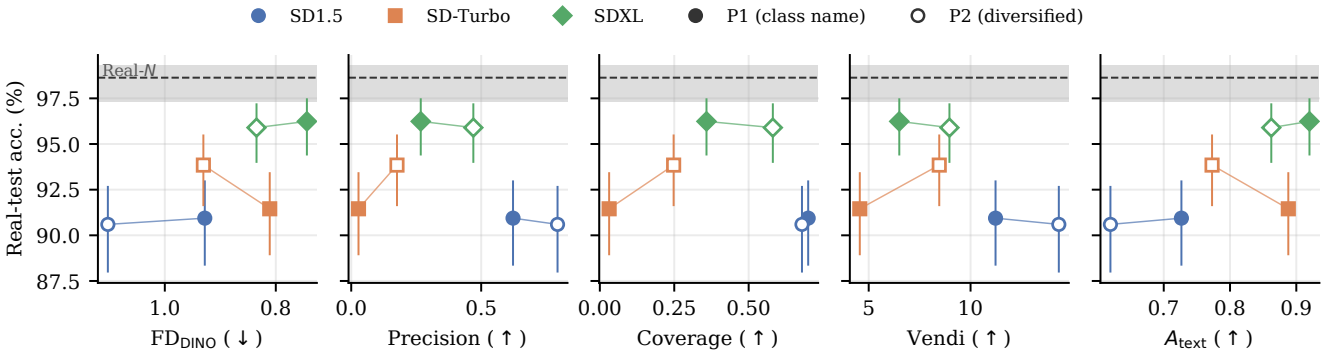


Figure 2. Which generator metric predicts training utility? Real-test accuracy against five generator-level metrics for three generators under P1 (filled) and P2 (open); bars are Wilson 95% intervals, the band is Real-N. FD_{DINO} (axis inverted) and A_{text} order the P1 generators correctly; precision, coverage and Vendi do not.

Training data (480 images)	Prompt	Real-test acc. (%) ↑	FD_{DINO} ↓	Precision ↑	Density ↑	Coverage ↑	Vendi ↑	A_{text} ↑
Real, 5/class	–	97.9 [96.4,98.8]	–	–	–	–	–	–
Real, 10/class	–	97.6 [96.0,98.6]	–	–	–	–	–	–
Real, all (≈24/class)	–	98.6 [97.3,99.3]	–	–	–	–	–	–
SD1.5	P1	90.9 [88.3,93.0]	0.93	0.62	0.04	0.70	11.2	0.73
	P2	90.6 [88.0,92.7]	1.10	0.79	0.03	0.68	14.3	0.62
SD-Turbo	P1	91.5 [88.9,93.5]	0.81	0.03	0.10	0.03	4.6	0.89
	P2	93.8 [91.6,95.5]	0.93	0.18	0.05	0.25	8.5	0.77
SDXL (768^2)	P1	96.2 [94.4,97.5]	0.74	0.27	0.11	0.36	6.5	0.92
	P2	95.9 [94.0,97.2]	0.83	0.47	0.07	0.58	9.0	0.86
VAE recon. (SD1.5)	–	98.3 [96.9,99.1]	0.11	0.98	0.88	1.00	5.2	–
VAE recon. (SD-Turbo)	–	98.5 [97.1,99.2]	0.11	0.98	0.91	1.00	5.2	–

Table 1. Cross-generator picture (20 CUB classes, frozen DINOv2 probe, random selection, one generation seed). Brackets: Wilson 95% intervals over the 585 test images. Bold: best synthetic value per column. Generator metrics are per-class averages in DINOv2 space. Precision ranks SD1.5, the worst teacher, first; only FD_{DINO} orders the three generators as their accuracy does. VAE recon.: real training images re-rendered through the generator's autoencoder (not run for SDXL).

		Effect on	Manipulation check			
Gen.	Factor	acc. (pp)	ΔVendi	ΔCov.	ΔPrec.	Inter.
SD1.5	diversity	+0.1	+1.50	+0.039	+0.089	+1.6
	alignment	+12.2	−4.33	+0.199	−0.023	
SD-Turbo	diversity	−0.7	+1.42	+0.108	+0.102	+0.3
	alignment	+3.9	−2.60	−0.023	+0.042	
SDXL	diversity	−0.5	+1.37	+0.127	+0.137	+0.9
	alignment	+1.3	−3.05	−0.004	−0.120	

Table 2. 2×2 factorial (P2): main effects and what each arm actually changed. Effects average over the other factor's levels; "Inter." is the diversity×alignment interaction. Raising alignment lowers Vendi by 2.6–4.3 yet raises accuracy; raising diversity lifts Vendi by ≈1.4 and coverage, but not accuracy.

tween -0.5 and +0.9 points with no consistent direction; real-only accuracy is already about 98%, so we cannot identify where synthetic data stops helping.

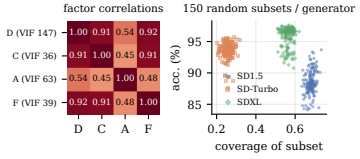


Figure 3. Why attribution fails at pilot scale. Left: correlations and variance-inflation factors of the four factors across 450 random subsets; D, C and F are nearly collinear. Right: subsets vary mostly between pools; within a pool, coverage moves little.

7. Discussion and limitations

What the interventions do and do not identify. Selecting from a fixed pool removes confounds that cross-generator comparisons cannot: architecture, autoen-



Figure 4. Where the gap lives. Per-class accuracy gap (real minus synthetic, P1) against class familiarity, with OLS fits. Most classes have no gap; two account for 58–82% of it. SD-Turbo's Bay-breasted Warbler has a 30-point gap at familiarity 0.995.

Method	Real /cls	SD1.5 Acc	SD1.5 Δ	SD-Turbo Acc	SD-Turbo Δ	SDXL Acc	SDXL Δ
Random-N	0	90.6	0	93.8	0	95.9	0
All-mN (3× images)	0	90.4	−0.02	95.4	+0.32	96.2	+0.12
k-center only	0	88.5	−0.26	92.5	−0.29	96.6	+0.25
Filter only (A_{text})	0	91.6	+0.13	96.2	+0.50	97.3	+0.50
DAS-text	0	94.0*	+0.43	95.0	+0.25	97.1	+0.44
Filter only (A_{real})	5	90.6	+0.00	95.2	+0.29	97.3	+0.50
DAS-real	5	93.0	+0.30	95.9	+0.43	97.1	+0.44
Real 5-shot + DAS-real	5	97.9	+0.91	98.5	+0.96	97.6	+0.62
Real 5-shot (no synthetic)	5			97.8			
Real-N (reference)	all			98.6			
DAS-real ablations (accuracy)							
τ = 20th pct.	5	93.2		95.0		96.4	
τ = 40th pct. (default)	5	93.0		95.9		97.1	
τ = 60th pct.	5	93.3		96.1		97.4	
centroid-nearest anchor	5	94.4		95.7		97.3	

Table 3. DAS versus its ablations (P2, 480 images unless noted). Δ: fraction of gap closed (Eq. 2). Bold: best synthetic-only set. *: $p<0.05$ vs. Random-N under a conservative unpaired test; one test image is 0.17 points. Real 5-shot was re-trained in this stage and differs from Tab. 1 by one test image.

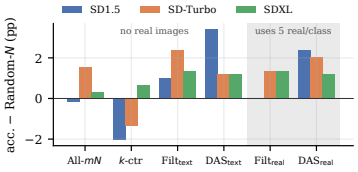


Figure 5. Gain over Random-N (480 images, P2). k-center alone hurts weaker generators; filtering by alignment carries most of DAS's benefit.



Figure 6. What the filter removes. The six lowest-A_{real} SD1.5 (P2) images for three classes chosen by a fixed random rule (rows are not hand-picked). Failures include a bird too small to identify, anatomically wrong puffins, and a different species. The top row also shows the cost of a 5-shot scorer: several plausible sparrows score low.

coder, training data and sampler. It does not make the four measured factors exhaustive. Any unmeasured property that co-varies with our selection rules is absorbed into the factor it travels with. Examples are texture statistics, which the predecessor found degraded [1], or background clutter. The decomposition says which measured property, when manipulated, moves accuracy; it is not a complete causal account of why a pool looks the way it does. The autoencoder ceiling bounds one unmeasured cause; others remain.

Scope of this version (what was not run). (i) Only 20 of 200 CUB classes, evenly spaced; they are easy (real data reaches 98.6%), which compresses differences. (ii) Three generators, one generation seed per pool, 585 test images: one image is 0.17 points, so many differences in Tab. 3 are one to six images. (iii) SDXL at 768^2; generators differ in steps and guidance as well as quality, so they are confounded. SD3.5 and FLUX, which the title question is really about, were not run. (iv) Not run: FGVC-Aircraft, HAM10000, P3, extra generation seeds, the from-scratch control, the CLIP-space and A_{text} reruns, the m and embedding ablations and SDXL's autoencoder ceiling; 150 instead of 500 subsets were drawn, and the Shapley shares have no confidence intervals. (v) SD-Turbo's P1 precision and coverage (0.03, 0.03) are unusually low and should be checked before being relied on.

Scope of the learner and data. The probe measures synthetic images as supervision for a head on a strong real-pretrained backbone, not as pretraining data. We cover one fine-grained domain, three open generators and classification only; generators fine-tuned on in-domain data [16] are out of scope. DAS-real needs five real images per class; DAS-text needs none, but its scorer inherits CLIP's weaknesses and may discard correct images on domains CLIP knows poorly.

Implications. For a weak generator, check label alignment before diversity; and do not rank generators by precision alone.

Data, licences and societal impact. We use only CUB-200-2011 under its research-use terms, no new human-subject data, and generators under their licences. The main risk is over-confidence: narrowing the benchmark gap could encourage replacing real with synthetic data in high-stakes domains; validate on real data there. Code and selection indices will be released.

LLM usage. [Authors to confirm.] An LLM assistant (Claude, Anthropic) drafted text, code and bibliography from the authors' plan; P2 templates: [authors:

LLM name/version]. [Authors: verify every citation and claim.] All numbers are written by code from stored outputs.

References

[1] Krzysztof Adamkiewicz, Brian B. Moser, Stanislav Frolov, Tobias Christian Nauen, Federico Raue, and Andreas Dengel. When pretty isn't useful: Investigating why modern text-to-image models fail as reliable training data generators. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 36660–36669, 2026. 2, 3, 4, 5, 8

[2] Shekoofeh Azizi, Simon Kornblith, Chitwan Saharia, Mohammad Norouzi, and David J. Fleet. Synthetic data from diffusion models improves ImageNet classification. Transactions on Machine Learning Research, 2023. 2

[3] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In ICLR, 2018. 3

[4] Black Forest Labs. FLUX.1 [dev]: A 12-billion-parameter text-to-image model. Model release, 2024. 3

[5] Lisa Dunlap, Alyssa Umino, Han Zhang, Jiezhi Yang, Joseph E. Gonzalez, and Trevor Darrell. Diversify your vision datasets with automatic diffusion-based augmentation. In NeurIPS, 2023. 3

[6] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In ICML, 2024. 3

[7] Lijie Fan, Kaifeng Chen, Dilip Krishnan, Dina Katabi, Phillip Isola, and Yonglong Tian. Scaling laws of synthetic images for model training …for now. In CVPR, 2024. 2

[8] Dan Friedman and Adji Bousso Dieng. The Vendi score: A diversity evaluation metric for machine learning. Transactions on Machine Learning Research, 2023. 2, 3

[9] Scott Geng, Cheng-Yu Hsieh, Vivek Ramanujan, Matthew Wallingford, Chun-Liang Li, Pang Wei Koh, and Ranjay Krishna. The unmet promise of synthetic training images: Using retrieved real images performs better. In NeurIPS, 2024. 2

[10] Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. Theoretical Computer Science, 38:293–306, 1985. 2, 3, 4, 5

[11] Ulrike Grömping. Relative importance for linear regression in R: The package relaimpo. Journal of Statistical Software, 17(1), 2006. 2, 5

[12] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In CVPR, 2016. 4

[13] Ruifei He, Shuyang Sun, Xin Yu, Chuhui Xue, Wenqing Zhang, Philip Torr, Song Bai, and Xiaojuan Qi. Is synthetic data from generative models ready for image recognition? In ICLR, 2023. 2

[14] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In NeurIPS, 2017. 2, 3

[15] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. arXiv preprint arXiv:2207.12598, 2022. 3

[16] Jae Myung Kim, Jessica Bader, Stephan Alaniz, Cordelia Schmid, and Zeynep Akata. DataDream: Few-shot guided dataset generation. In ECCV, 2024. 3, 8

[17] Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do better ImageNet models transfer better? In CVPR, 2019. 4

[18] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. In NeurIPS, 2019. 2, 3, 4

[19] Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of ImageNet classes in Fréchet inception distance. In ICLR, 2023. 2, 3

[20] Shiye Lei, Hao Chen, Sen Zhang, Bo Zhao, and Dacheng Tao. Image captions are natural prompts for text-to-image models. arXiv preprint, 2023. 3

[21] Stan Lipovetsky and Michael Conklin. Analysis of regression in game theory approach. Applied Stochastic Models in Business and Industry, 17(4):319–330, 2001. 2, 5

[22] Subhransu Maji, Esa Rahtu, Juho Kannala, Matthew Blaschko, and Andrea Vedaldi. Fine-grained visual classification of aircraft. arXiv preprint arXiv:1306.5151, 2013. 3

[23] Eyal Michaeli and Ohad Fried. Advancing fine-grained classification by structure and subject preserving augmentation. In NeurIPS, 2024. 3

[24] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In ICML, 2020. 2, 3, 4

[25] Curtis Northcutt, Lu Jiang, and Isaac Chuang. Confident learning: Estimating uncertainty in dataset labels. Journal of Artificial Intelligence Research, 70: 1373–1411, 2021. 3

[26] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research, 2024. 3

[27] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image synthesis. In ICLR, 2024. 3

[28] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In ICML, 2021. 3

[29] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In CVPR, 2022. 2, 3

[30] Mert Bülent Sarıyıldız, Karteek Alahari, Diane Larlus, and Yannis Kalantidis. Fake it till you make it: Learning transferable representations from synthetic ImageNet clones. In CVPR, pages 8011–8021, 2023. 2, 3

[31] Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In ECCV, 2024. 3

[32] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In ICLR, 2018. 2, 3, 5

[33] Jordan Shipard, Arnold Wiliem, Kien Nguyen Thanh, Wei Xiang, and Clinton Fookes. Diversity is definitely needed: Improving model-agnostic zero-shot classification via stable diffusion. In CVPR Workshops, 2023. 3

[34] Krishnakant Singh, Thanush Navaratnam, Jannik Holmer, Simone Schaub-Meyer, and Stefan Roth. Is synthetic data all we need? benchmarking the robustness of models trained with synthetic images. In CVPR Workshops, pages 2505–2515, 2024. 2

[35] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. Beyond neural scaling laws: Beating power law scaling via data pruning. In NeurIPS, 2022. 3

[36] Stability AI. Introducing Stable Diffusion 3.5. Model release, 2024. 3

[37] George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In NeurIPS, 2023. 3, 4

[38] Yonglong Tian, Lijie Fan, Phillip Isola, Huiwen Chang, and Dilip Krishnan. StableRep: Synthetic images from text-to-image models make strong visual representation learners. In NeurIPS, 2023. 2

[39] Yonglong Tian, Lijie Fan, Kaifeng Chen, Dina Katabi, Dilip Krishnan, and Phillip Isola. Learning vision from models rivals learning vision from data. In CVPR, 2024. 2

[40] Brandon Trabucco, Kyle Doherty, Max Gurinas, and Ruslan Salakhutdinov. Effective data augmentation with diffusion models. In ICLR, 2024. 3

[41] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Scientific Data, 5:180161, 2018. 3

[42] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The Caltech-UCSD Birds-200-2011 dataset. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011. 3

[43] Jianhao Yuan, Jie Zhang, Shuyang Sun, Philip Torr, and Bo Zhao. Real-fake: Effective training data synthesis through distribution matching. In ICLR, 2024. 3